Deepika – Medicine Search & Availability Module

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Module 3 of 4, owned by Deepika: finding nearby pharmacies with enough stock, comparing them, contacting them and navigating to one.

1. Module description and objectives

Deepika's module lets a patient or hospital find, within seconds, which nearby approved pharmacies hold the needed quantity of a medicine, compare them on price, distance and delivery, contact one directly, and navigate to it with Google Maps when delivery is not possible. It reads data from Divya's, Leelavathy's and Anusuya's modules through interfaces and owns only two tables: search history and contact logs.

Technology: HTML, CSS, JavaScript, Bootstrap 5 (frontend); Java 17 with Spring Boot 3, Spring Data JPA, Maven (backend); MySQL 8 with Flyway migrations (database); Google Maps Directions URL for navigation.

Objectives

- Search medicines by name and required quantity.
- List only pharmacies with sufficient, unexpired stock, with price, quantity, distance, address and delivery availability.
- Sort and compare pharmacies.
- Provide Call and WhatsApp contact options and Google Maps navigation.
- Suggest alternatives when stock is unavailable.
- Hand the user over to Anusuya's request form with the choice already made.

2. Features

- Medicine typeahead (uses Leelavathy's `GET /medicines?q=`) and quantity input.
- Results show pharmacy name, address, distance in km, approximate travel time, available quantity, unit price, total price, stock badge, Demo or Live badge with last-updated time, open or closed now, delivery status and fee, and a prescription warning.
- Sort by distance, price, quantity or delivery first; filter by radius and Delivery only.
- Compare up to three pharmacies side by side.
- Contact: Call (`tel:`) and WhatsApp (`wa.me` with a prefilled message), both logged.

- Location from browser geolocation, saved profile location or manual coordinates; distance by Haversine.
- Navigation: Google Maps Directions URL from the user's location to the pharmacy (no API key required).
- Alternatives: wider radius results, partial stock and same-generic substitutes (marked Confirm with a doctor or pharmacist).
- Recent searches (kept 30 days) and a disclaimer banner that demo data is not real-time stock.

3. Workflow

1. The user opens `search.html`. The page asks for browser location, then falls back to the saved profile location, then to manual entry.
2. The user picks a medicine, enters quantity and radius (default 10 km) and presses Search.
3. `GET /search/pharmacies` finds nearby approved pharmacies (Divya's `PharmacyLookup`), their offers (Leelavathy's `InventoryQueryService`), settings and delivery status (Leelavathy's `PharmacySettingsLookup` and Anusuya's `DeliveryAvailabilityQuery`), then computes distance, sorts and logs the search with the location rounded to 3 decimals.
4. Result cards appear. A red banner warns when the medicine needs a prescription.
5. The user can tick up to three cards and press Compare.
6. On a card the user chooses:
   - Call or WhatsApp: the page logs the contact, then opens the link.
   - Request delivery: opens Anusuya's `request/new.html` with `fulfilment=DELIVERY` (shown only when delivery is available or busy).
   - Self-pickup and navigate: opens Google Maps with the route and opens `request/new.html` with `fulfilment=PICKUP` so stock is held. Highlighted when delivery is unavailable.
7. If nothing matches, the page shows an empty state with a wider-radius search, partial stock and substitutes (HTTP 200, not an error).

4. Frontend requirements

Use HTML, CSS, JavaScript and Bootstrap 5.3. Reuse Divya's `theme.css`, `api.js`, `ui.js` and `auth-guard.js`: Navy `#102A43` primary, Teal `#0D9488` secondary, Red `#DC2626` only for the emergency Request button and the prescription warning, backgrounds `#F1F5F9`

and white, font Inter. Add only `search.css`, `search.js`, `compare.js` and `geo.js`. Every page needs a skeleton loader, empty state, inline validation and toasts.

Page	Purpose and active controls
`search.html`	Search form: medicine typeahead, quantity stepper, radius select, Delivery only switch, Sort select, Use my location button, manual coordinates in an Advanced panel. Search calls `GET /search/pharmacies`. Recent-search chips call `GET /search/history`; Clear calls `DELETE /search/history`. Result cards with Call, WhatsApp, Navigate, Request, Details and a Compare checkbox. Sticky compare tray (maximum 3) with a Compare button. Amber banner: demo data, confirm stock by phone before travelling.
`pharmacy-details.html`	Pharmacy header, hours, contact buttons, delivery info, the offer for the searched medicine, Navigate, Back to results. Calls `GET /pharmacies/{id}/details`.
`compare.html` (also a modal)	One column per pharmacy; rows for distance, price, total, quantity, expiry, delivery, fee, hours, data source; the best value in each row is highlighted teal. Calls `GET /search/compare`. A Request button under each column.

Card design: Navy headings, Teal primary actions. Delivery badge: green Available, amber Busy, grey Unavailable. Stock badge and Demo or Live badge on every card. A red banner appears when the medicine requires a prescription. Request delivery is shown only when delivery is Available or Busy; Self-pickup and navigate is always shown and highlighted when delivery is Unavailable. Location denied in the browser opens the manual-entry panel.

5. Spring Boot backend requirements

- Orchestrate the search through interfaces only: `PharmacyLookup.findApprovedWithin`, `InventoryQueryService.findOffers`, `PharmacySettingsLookup`, `MedicineLookup`, `DeliveryAvailabilityQuery`. No direct access to other modules' tables.

- Exact distance with `GeoUtils.haversineKm`; approximate travel time = distance divided by 25 km/h, labelled approx; `totalPrice = unitPrice x quantity`.
- Open-now calculation in Asia/Kolkata, including hours that pass midnight and `opens_24x7`.
- Build contact links on the server: `tel:+91<number>` and `https://wa.me/<digits>?text=<url-encoded message>` naming medicine and quantity only (no patient name or medical details).
- Build the Directions URL with `MapLinks`: `https://www.google.com/maps/dir/?api=1&origin=LAT,LNG&destination=LAT,LNG&travelmode=driving`. No API key is needed; if a Maps JavaScript embed is added, read the key from an environment variable and never commit it.
- Alternatives engine when there are no matches: radius times 2.5 (maximum 50 km), partial stock, and substitutes from `findAlternatives`.
- `DefaultDeliveryAvailabilityQuery` as a `@ConditionalOnMissingBean` fallback until Anusuya's module is merged. If the delivery lookup fails, treat the status as `UNAVAILABLE` and continue.
- Store `search_history` location rounded to 3 decimals; nightly job deletes rows older than 30 days.
- Flyway version `V300` only.

Sample search result item

```
{
  "pharmacyId": 1,
  "name": "MediCare Pharmacy",
  "address": "12 Lattice Bridge Rd, Adyar, Chennai",
  "distanceKm": 2.4,
  "etaMinutes": 6,
  "openNow": true,
  "inventoryId": 41,
  "availableQuantity": 35,
  "unitPrice": 28.50,
  "totalPrice": 114.00,
  "stockStatus": "IN_STOCK",
  "dataSource": "SAMPLE",
  "lastUpdatedAt": "2026-10-09T10:15:00Z",
  "requiresPrescription": false,
  "delivery": { "status": "AVAILABLE", "fee": 30.00, "radiusKm": 8.0 },
  "contact": { "callUrl": "tel:+919000000001", "whatsappUrl":
"https://wa.me/919000000001?text=..." },
  "navigationUrl": "https://www.google.com/maps/dir/?
api=1&origin=13.04,80.23&destination=13.0012,80.2565&travelmode=driving"
}
```


```
com.emergencymed.search
  controller/ SearchController, PharmacyDetailsController, ContactController,
NavigationController
  service/    PharmacySearchService, ComparisonService,
AlternativeSuggestionService,
              ContactLinkService, NavigationService, SearchHistoryService,
OpeningHoursService
  repository/ SearchHistoryRepository, ContactLogRepository
  entity/     SearchHistory, ContactLog
  dto/        SearchRequest, PharmacyMatch, DeliveryInfo, ContactInfo,
AlternativesResponse, CompareResponse
  fallback/   DefaultDeliveryAvailabilityQuery   (@ConditionalOnMissingBean)
  config/     MapsProperties (maps.directions.base-url, maps.api-key from
environment)
  job/        SearchHistoryCleanupJob
src/main/resources/db/migration  V300__m3_search_contact_logs.sql
src/main/resources/static/       search.html, pharmacy-details.html,
compare.html
src/main/resources/static/assets/ css/search.css, js/search.js,
js/compare.js, js/geo.js
```

6. MySQL database tables (owned by Deepika)

Two tables, Flyway file `V300__m3_search_contact_logs.sql`, created after Divya's and Leelavathy's migrations. Everything else the search needs is read through interfaces from tables owned by the other members.

`search_history`: `id` PK; `user_id` FK to `users.id`; `medicine_id` FK to `medicines.id` NULL; `query_text` VARCHAR(120); `quantity` INT; `user_lat`, `user_lng` DECIMAL(6,3) (rounded for privacy); `radius_km` DECIMAL(4,1); `sort_by` VARCHAR(15); `results_count` INT; `created_at` DATETIME. Index (`user_id`, `created_at`).

`contact_logs`: `id` PK; `user_id` FK to `users.id`; `pharmacy_id` FK to `pharmacies.id`; `medicine_id` FK to `medicines.id` NULL; `contact_type` VARCHAR(10) (`CALL`, `WHATSAPP`, `NAVIGATION`); `created_at` DATETIME. Index (`pharmacy_id`, `created_at`).

Relationships: user 1 to many searches; user 1 to many contact logs; pharmacy 1 to many contact logs; medicine 1 to many searches.

Data read through interfaces (not owned): pharmacies and coordinates (`PharmacyLookup`, Divya), medicines and prescription flag (`MedicineLookup`, Leelavathy), stock and price (`InventoryQueryService`, Leelavathy), contact and delivery settings (`PharmacySettingsLookup`, Leelavathy), live delivery availability (`DeliveryAvailabilityQuery`, Anusuya).

7. REST APIs

Base path `/api/v1`, shared response envelope. Roles: `PATIENT`, `HOSPITAL`, `ADMIN`.

Method	Path	Purpose
GET	`/search/pharmacies`	Params `medicineId`, `quantity`, `lat`, `lng`, `radiusKm` (1 to 50, default 10), `sort` (`distance`, `price`, `quantity`, `delivery`), `deliveryOnly`, `page`, `size`. Returns `matches`; when empty also returns `alternatives`.
GET	`/search/compare`	Params `pharmacyIds` (2 to 3), `medicineId`, `quantity`, `lat`, `lng`
GET	`/search/history`	Last 10 searches of the current user
DELETE	`/search/history`	Clear own history
GET	`/pharmacies/{id}/details`	Pharmacy plus offer; params `medicineId`, `quantity`, `lat`, `lng`

Method	Path	Purpose
GET	`/contact/{pharmacyId}/links`	Returns `callUrl` and `whatsappUrl`; params `medicineId`, `quantity`
POST	`/contact/log`	Body `{pharmacyId, medicineId?, contactType}`
GET	`/navigation/{pharmacyId}`	Params `lat`, `lng`, `mode`; returns `directionsUrl` and logs `NAVIGATION`

Medicine typeahead uses Leelavathy's `GET /medicines?q=`; it is not duplicated here. Anusuya's track page also calls `GET /contact/{pharmacyId}/links` and `GET /navigation/{pharmacyId}`, so these two endpoints must also accept the requester of an existing request (roles `PATIENT` and `HOSPITAL`, already allowed).

8. Integration requirements

What Deepika provides and consumes

- Provides: the search, compare, details, contact-link and navigation endpoints. Anusuya's track page calls `GET /contact/{pharmacyId}/links` and `GET /navigation/{pharmacyId}`; her request form is opened from this module with URL parameters.
- Consumes from Divya: `CurrentUserProvider`, `PharmacyLookup.findApprovedWithin`, `GeoUtils`, `MapLinks`, `PhoneUtils`, and the saved location through `GET /users/me`.
- Consumes from Leelavathy: `MedicineLookup`, `InventoryQueryService` (`findOffers`, `findAlternatives`), `PharmacySettingsLookup`, and `GET /medicines?q=` for the typeahead.
- Consumes from Anusuya: `DeliveryAvailabilityQuery` (with her `DefaultDeliveryAvailabilityQuery` fallback until merged).
- Page hand-off to Anusuya: `/request/new.html?pharmacyId=7&medicineId=3&quantity=4&fulfilment=PICKUP` (or `DELIVERY`). Hand-off back: `/pharmacy-details.html?id=7`. Leelavathy's catalogue opens `/search.html?medicineId=ID`.

Shared conventions (identical in all four module documents)

- Base path `/api/v1`. Every response is `{success, message, data, errors[{field,message}], timestamp}`. Lists use `data = {content, page, size, totalElements, totalPages}` with params `page` (from 0), `size` (default 10, maximum 50) and `sort`.

- Roles: `PATIENT`, `HOSPITAL`, `PHARMACY`, `DELIVERY`, `ADMIN`. The user and pharmacy ID always come from the session through `CurrentUserProvider`, never from the request body.
- Naming: tables plural `snake_case`, entities singular `PascalCase`, JSON `camelCase`, enums `UPPER_SNAKE` stored as strings, `BIGINT` IDs, money `DECIMAL(10,2)` in INR, coordinates `DECIMAL(9,6)`, timestamps in UTC.
- Login is a server-side session; every `POST`, `PUT`, `PATCH`, `DELETE` carries the CSRF header `X-XSRF-TOKEN` (added by `api.js`).
- Stock: `availableQuantity = quantity - reservedQuantity`. Inventory rows are `SAMPLE` or `LIVE`; sample data is never presented as real-time stock.
- Prescription-only medicines need a prescription verified by the pharmacy before a request is accepted.

HTTP	`errorCode`	When
400	`VALIDATION_ERROR`	Bean validation or bad parameter
401	`UNAUTHORIZED`	Not logged in or session expired
403	`FORBIDDEN`	Wrong role or another user's data
403	`PHARMACY_NOT_APPROVED`	Pharmacy action before approval
404	`NOT_FOUND`	Missing record
409	`DUPLICATE_RESOURCE`	Email, licence number or medicine already exists
409	`INSUFFICIENT_STOCK`	Reservation larger than available stock
409	`INVALID_STATE_TRANSITION`	Status change not allowed
409	`NO_DELIVERY_PARTNER`	Delivery not possible; offer self-pickup
422	`PRESCRIPTION_REQUIRED`	Prescription-only item without a verified prescription
500	`INTERNAL_ERROR`	Unexpected; generic message only

Shared database design (one MySQL database `emergency_med_db`, 19 tables)

Owner	Module	Tables	Flyway range
Divya	User & Authentication	`users`, `patient_profiles`, `hospital_profiles`, `pharmacies`, `delivery_partner_profiles`	`V100` to `V199`

Owner	Module	Tables	Flyway range
Leelavathy	Medicine & Inventory	`medicine_categories`, `medicines`, `pharmacy_settings`, `pharmacy_inventory`, `inventory_transactions`	`V200` to `V299`
Deepika	Medicine Search & Availability	`search_history`, `contact_logs`	`V300` to `V399`
Anusuya	Emergency Request & Notification	`emergency_requests`, `request_items`, `prescriptions`, `request_status_history`, `delivery_availability`, `delivery_assignments`, `notifications`	`V400` to `V499`

Only the owner reads or writes a table directly. Deepika's module reads the other tables only through the interfaces below and never queries them. No `ON DELETE CASCADE` across modules.

Interface (`com.emergencymed.common.contract`)	Implemented by	Used by	Methods
`CurrentUserProvider`	Divya	all	`Long userId()`, `Role role()`, `Optional<Long> pharmacyId()`, `boolean isAuthenticated()`
`PharmacyLookup`	Divya	Leelavathy, Deepika, Anusuya	`findApproved(id)`, `findApprovedWithin(lat, lng, radiusKm)`, `pharmacyIdForUser(userId)`
`UserLookup`	Divya	Anusuya	`findById(userId)`, `activeDeliveryPartnerIds()`
`MedicineLookup`	Leelavathy	Deepika, Anusuya	`findById(id)`, `searchByName(q, limit)`
`InventoryQueryService`	Leelavathy	Deepika	`findOffers(medicineId, qty, pharmacyIds)`, `findAlternatives(medicineId, pharmacyIds)`
`InventoryService`	Leelavathy	Anusuya	`reserve`, `commit`, `release`, `restore`
`PharmacySettingsLookup`	Leelavathy	Deepika, Anusuya	`find(pharmacyId)`

Interface (`com.emergencymed.common.contract`)	Implemented by	Used by	Methods
`DeliveryAvailabilityQuery`	Anusuya	Deepika	`statusFor(pharmacyId)`: `AVAILABLE`, `BUSY`, `UNAVAILABLE`
`NotificationPublisher`	Anusuya	Divya, Leelavathy, Anusuya	`publish(userId, type, title, message, requestId)`

Every interface Deepika consumes needs a `@ConditionalOnMissingBean` fallback so her module runs alone before integration.

9. Validation, exception handling and security

Input	Rule
`medicineId`	Required, must exist and be active; otherwise 404 `NOT_FOUND`
`quantity`	Integer 1 to 1000
`lat`, `lng`	Required, in range; missing returns 400 with the message Location is required
`radiusKm`	1 to 50, default 10
`sort`	One of `distance`, `price`, `quantity`, `delivery`, otherwise 400
`pharmacyIds` for compare	2 to 3 distinct IDs, all approved
`contactType`	One of `CALL`, `WHATSAPP`, `NAVIGATION`

- Zero matches is a normal 200 response with an `alternatives` block, not an error.
- If Anusuya's module is down or slow, delivery status falls back to `UNAVAILABLE` and the search still succeeds.
- Pharmacy phone numbers are returned only to logged-in users. The WhatsApp text is URL-encoded and contains no patient name or medical history.
- Only `PATIENT`, `HOSPITAL` and `ADMIN` can search; other roles get 403.
- Search location is stored rounded to 3 decimals; history is deleted after 30 days.
- Optional soft rate limit of 30 searches per minute per user (HTTP 429 `RATE_LIMITED`).
- Any Maps API key comes from an environment variable and is never committed.

10. Testing scenarios and expected output

Scenario	Steps	Expected
Basic search	Paracetamol 650 mg, quantity 5, radius 10 km from Adyar	At least two cards, sorted by distance, each with distance and price
Quantity filter	Search quantity 200	Pharmacies with less stock are excluded; alternatives show partial stock
Out of stock	Search a medicine with all rows at 0	Empty state with wider-radius and substitute suggestions; HTTP 200
Sort by price	Choose Price	Cards ascending by `unitPrice`
Expired batch	A row whose expiry was yesterday	Never listed
Unapproved pharmacy	`pharmacy.new@demo.in` has stock	Never listed
Delivery unavailable	Pharmacy with delivery off	Grey badge; Self-pickup and navigate highlighted; no Request delivery button
Navigation	Press Navigate	New tab opens Google Maps with origin and destination filled; `NAVIGATION` logged
WhatsApp	Press WhatsApp	Chat opens with a prefilled message naming medicine and quantity; `WHATSAPP` logged
Prescription item	Search Adrenaline injection	Red prescription banner on every card
Compare	Tick 3 cards, Compare	Table with best values highlighted; a 4th tick is blocked with a message
Location denied	Block browser location	Manual entry panel opens; search works with typed coordinates
Bad input	Quantity 0	400 `VALIDATION_ERROR`

Scenario	Steps	Expected
Open now	Pharmacy hours 22:00 to 06:00 checked at 01:00 IST	`openNow = true`
Distance	Adyar to T. Nagar	About 5 km